

Statement of Work (SOW)

Project Title: Elective Course Registration System

1. Introduction

This Statement of Work (SOW) outlines the objectives, scope, deliverables, and responsibilities for the Elective Course Registration System to be executed by .

2. Scope of Work

Project Description:

This project aims to develop a web-based Elective Course System for Mahindra University students, designed to streamline the process of selecting elective courses while accommodating fixed mandatory courses. The system provides a visual timetable interface that helps students plan their schedules efficiently which includes built-in validation and conflict detection to prevent overlapping courses, this also removes the need of back-up courses. It incorporates an approval mechanism to handle special cases and eliminate manual errors, reduce administrative workload, and ensure a smooth, transparent registration experience. Overall, the system will enhance efficiency, accuracy, and student satisfaction during the academic registration process.

Objectives:

1. To design and develop a user-friendly elective/course registration system with a visual timetable interface.
2. To enable students to select elective courses while simultaneously checking for clashes and automatically accommodating fixed mandatory courses.
3. To implement an approval workflow for special cases and exception handling by making a secure role-based authentication (Student, Admin?).
4. Implement automated prerequisite validation and enforce credit limit constraints.
5. Ensure mobile compatibility and accessibility for better and easy use.
6. Maintain data integrity using PostgreSQL constraints.
7. Provide analytics dashboard for administrators.

Key Activities:

- Requirements gathering and Analysis
- System Architecture Design (3-tier architecture: frontend, backend and DB)
- Database Design (ER Diagram and Schema Creation)
- API Design and Documentation (REST APIs using Spring Boot)
- Frontend Development using React, Backend Development using Spring Boot, Integration of Frontend and Backend
- Implementation of Validation and Conflict Detection Logic
- Testing, Performance Optimization and Debugging
- Deployment and Documentation Preparation

3. Deliverables

TBD

4. Timeline and Milestones

Milestone	Description	Due Date	Progress
Initial Team Meeting	Discuss project overview, ideas/features and possible plan of action with SOW.	06/02/26	DONE
Weekly Team meet with mentor in PWS	Talk to sir about the plan of action and SOW, get advice on how to proceed.	09/02/26	DONE
Meet with Haritha Ma'am	Discuss about admin and student troubles that she has been facing with respect to elective allocations.	09/02/26	DONE
Talk to Avinash sir	Clear doubts about the project.	11/02/26	
Frontend (React)	Try completing the majority of front end design planning and execution.	10/03/26	
Backend (SpringBoot)			
Database (PostgreSQL)			

5. Roles and Responsibilities

To be discussed after the scale of the project is fully fixed by meeting with professor. Client

Contact: Software Engineering course, Mahindra University

6. Budget and Payment Terms (For now you can ignore this)

7. Assumptions and Constraints

Assumptions:

1. The timetable, faculty assignments, class days, and time slots are predefined and will not change during the registration period.
2. The system closely follows Mahindra University's existing class structure, academic days, and scheduling format.
3. Course capacities, prerequisites, and eligibility rules are correctly defined before registration begins.
4. Students' academic records used by the system are accurate and up to date.

Constraints:

1. Course allocation follows a first come, first served approach rather than CGPA.
2. CGPA is only used to resolve ties when multiple students register at the same time.
3. Students are only shown courses they are eligible for based on year, branch, prerequisites, and timetable constraints; therefore, backup course selection is not required.
4. The system is limited to selected schools and branches and does not cover the entire university.

8. Approval Signatures

Client:

Name: _____

Title: _____

Signature: _____

Date: _____

Service Provider:

Name: _____

Title: _____

Signature: _____

Date: _____